

FORECASTING THE FED'S NEXT MOVE

THE AGENT MODEL APPROACH

Prior Research vs. Enhanced vs. Agent Strategy

PART 1: INTRODUCTION

Purpose & Importance

RESEARCH PURPOSE

Goal: Reliably forecast U.S. Fed interest rate decisions (Raise, Hold, Lower) using a Multi-Modal approach.

- **Problem:** Traditional models miss nuanced textual signals.
- **Solution:** Combine Macroeconomic Indicators + Textual Sentiment (Fed Communications).

PART 2: PRIOR RESEARCH

Method 1 (The Baseline)

METHOD 1: BASELINE ARCHITECTURE

XGBoost Classifier trained on:

1. ECONOMIC DATA

Standardized indicators (CPI, Unemployment, etc.)

2. TEXT FEATURES

TF-IDF Vectors + LM Sentiment Scores.

Performance: AUC 0.8535. Good, but struggled with minority classes (Rate Cuts).

PART 3: ENHANCED METHODOLOGY

Improving Feature Engineering

KEY IMPROVEMENTS

- **Text:** Added Bi-grams (e.g., "inflation target") to capture context.
- **Data:** Added 3M/6M Rolling Averages to capture economic *trends*.
- **Tuning:** RandomizedSearchCV for dynamic hyperparameter optimization.
Result: AUC improved to 0.8671.

PART 4: THE AGENT MODEL STRATEGY

A Stacking Ensemble Approach

THE "AGENT" CONCEPT

Instead of relying on one model, we built a **Stacking Ensemble**.

LEVEL 1: SPECIALIST AGENTS

Diverse models trained separately:

- Logistic Regression (The Conservative)
- Random Forest (The Generalist)
- XGBoost (The Trend Spotter)

LEVEL 2: DECISION AGENT

A Meta-Learner (Logistic Regression) that takes the *opinions* (probability predictions) of the specialists + economic data to make the final call.

MOTIVATION: WHY USE AGENTS?

1. DIVERSITY OF OPINION

Single models have blind spots. Linear models miss non-linear trends; Tree models can overfit. Agents cover each other's weaknesses.

2. ROBUSTNESS TO IMBALANCE

"Rate Cuts" are rare events. By stacking models, the Decision Agent learns to trust the specific specialist that is best at spotting these rare signals.

WHY DID IT PERFORM BETTER?

- **Superior Ranking:** The Agent model achieved an AUC of **0.8735**, the highest of all experiments.
- **Perfect Recall on "Cuts":**
Recall for Class -1 (Lower): 1.00
Unlike previous models that ignored rate cuts, the Decision Agent successfully identified 100% of cut instances in the test set.
- **Trust Calibration:** The meta-learner learned to weigh the "confidence" of each agent, reducing false alarms on the "Hold" class.

FINAL RESULTS COMPARISON

Model Strategy	Test AUC	Key Strength
Prior Research (Method 1)	0.8535	Baseline
Enhanced XGBoost	0.8671	Trend Detection
Agent Model (Stacking)	0.8735	Recall on Rare Events

Conclusion: The Agent Model architecture provides the most robust and safety-critical predictions for monetary policy forecasting.

Speaker notes